

Multiple Choice

1. **Answer:** B

Options: A) ALDH 1; B) ALDH 2; C) ALDH 3; D) ALDH 4

Note: Facial flushing in approximately 50% of East Asians is associated with a deficiency of the enzyme ALDH 2, which impairs the metabolism of acetaldehyde, a byproduct of alcohol consumption, leading to uncomfortable physiological reactions.

2. **Answer:** C

Options: A) Superoxide dismutase; B) Alkaline phosphatase; C) Glutathione peroxidase; D) DNA polymerase

Note: Glutathione peroxidase is not a zinc-dependent enzyme because it primarily relies on selenium as a cofactor for its enzymatic activity, unlike other enzymes that require zinc for their function.

3. **Answer:** A

Options: A) Carries a risk of colonic cancer; B) Can be treated with a LOFFLEX diet; C) Can be caused by small intestinal bacterial overgrowth; D) Can be caused by milk intolerance

Note: Ulcerative colitis is a chronic inflammatory bowel disease that increases the risk of developing colonic cancer due to prolonged inflammation and dysplasia in the colon.

4. **Answer:** D

Options: A) Vitamin A; B) Folic acid; C) Vitamin C; D) Vitamin D

Note: Vitamin D is essential for calcium absorption and bone health, and its deficiency can lead to impaired enamel formation and a higher susceptibility to dental caries.

5. **Answer:** A

Options: A) Vegan diets are likely to be deficient in protein; B) Cereals are deficient in lysine; C) Legumes are deficient in methionine; D) Dairy products are rich source of tryptophan

Note: Vegan diets can provide sufficient protein through a variety of plant-based sources such as legumes, nuts, seeds, and whole grains, making the statement that they are likely to be deficient in protein inaccurate.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: Riboflavin is not the primary vitamin for fatty acid synthesis; instead, biotin plays a crucial role as a coenzyme in this process.

7. **Answer:** True

Options: A) True; B) False

Note: Potassium helps to balance sodium levels in the body, which can lead to lower blood pressure and improved hypertension outcomes in humans.

8. **Answer:** False

Options: A) True; B) False

Note: Acarbose is not considered a first-line medication for managing type 2 diabetes in obese patients; instead, metformin is typically the preferred initial treatment due to its effectiveness and safety profile.

9. **Answer:** True

Options: A) True; B) False

Note: Epidemiological measures of effect, such as risk ratios and odds ratios, quantify the relationship between an exposure and an outcome, thereby assessing the strength of that association.

10. **Answer:** False

Options: A) True; B) False

Note: The statement is false because starch primarily consists of amylose and amylopectin, which are polymers of glucose connected by alpha 1,4-linkages, not fructose and beta linkages.

Long Form Response

11. **Answer:** ATP, or adenosine triphosphate, serves as the immediate source of chemical energy required for skeletal muscle contraction. When a muscle cell is stimulated, ATP is produced primarily through cellular respiration, which involves breaking down glucose and fatty acids. During contraction, ATP binds to the myosin heads in muscle fibers, allowing them to pull on actin filaments, resulting in muscle shortening and movement. The significance of ATP in muscle function cannot be overstated, as it ensures that muscles can contract and relax efficiently, enabling physical activity and overall mobility. Without sufficient ATP, muscles would fatigue quickly and be unable to perform effectively.

Note: correct because it accurately describes ATP's essential role in muscle contraction by detailing its production through cellular respiration, its binding to myosin heads for actin interaction, and its critical importance in facilitating muscle movement.

12. **Answer:** Binge Eating Disorder (BED) can be effectively treated using various methods, including psychotherapy, such as cognitive-behavioral therapy (CBT), and medications like certain antidepressants. These treatments focus on helping individuals understand their eating patterns and develop healthier coping strategies. However, benzodiazepines, such as Diazepam and Alprazolam, have not shown effectiveness in managing BED. This is because benzodiazepines primarily address anxiety and are not designed to tackle the underlying behavioral and emotional issues associated with

binge eating. Therefore, they do not provide the necessary support to change eating behaviors or improve emotional regulation in individuals with BED.

Note: correct because it identifies effective treatments for Binge Eating Disorder, such as psychotherapy and certain antidepressants, while explaining that benzodiazepines are ineffective due to their primary function of managing anxiety rather than addressing the underlying issues related to eating behaviors.

End of Answer Key